Data Engineering

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Amazon Kinesis Data Streams



- Collect and store streaming data in **real-time**



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Kinesis Data Streams



- Retention between up to 365 days
- Ability to reprocess (replay) data by consumers
- Data can't be deleted from Kinesis (until it expires)
- Data up to 10MiB (typical use case is lot of "small" **real-time data**)
- Data ordering guarantee for data with the same "Partition ID"
- At-rest KMS encryption, in-flight HTTPS encryption
- Kinesis Producer Library (KPL) to write an optimized producer application
- Kinesis Client Library (KCL) to write an optimized consumer application

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Kinesis Data Streams – Capacity Modes

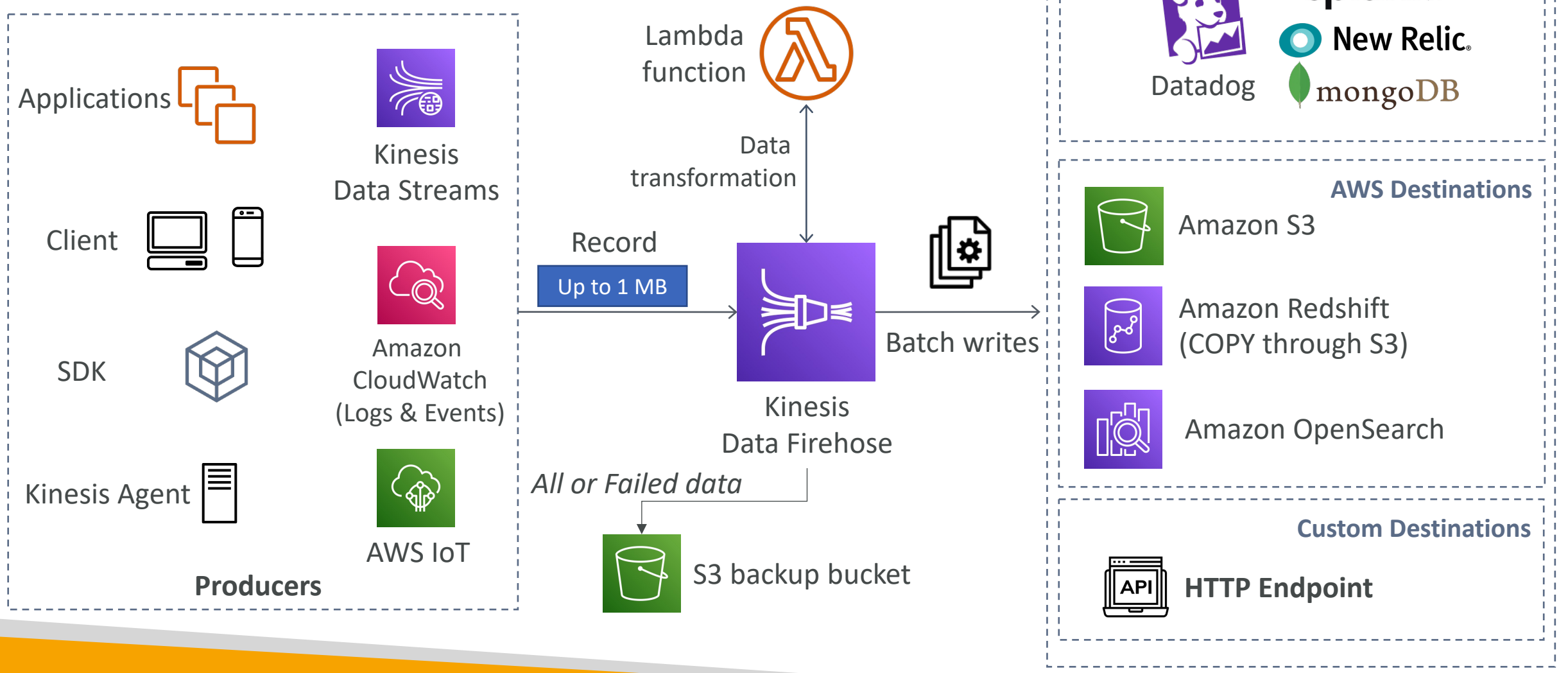
- Provisioned mode:
  - Choose number of shards
  - Each shard gets 1MB/s in (or 1000 records per second)
  - Each shard gets 2MB/s out
  - Scale manually to increase or decrease the number of shards
  - You pay per shard provisioned per hour

- On-demand mode:
  - No need to provision or manage the capacity
  - Default capacity provisioned (4 MB/s in or 4000 records per second)
  - Scales automatically based on observed throughput peak during the last 30 days
  - Pay per stream per hour & data in/out per GB

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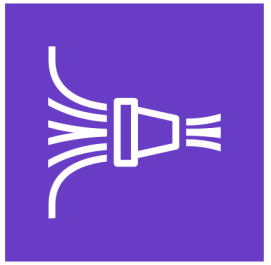
Kinesis Data Firehose



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AWS Kinesis Data Firehose

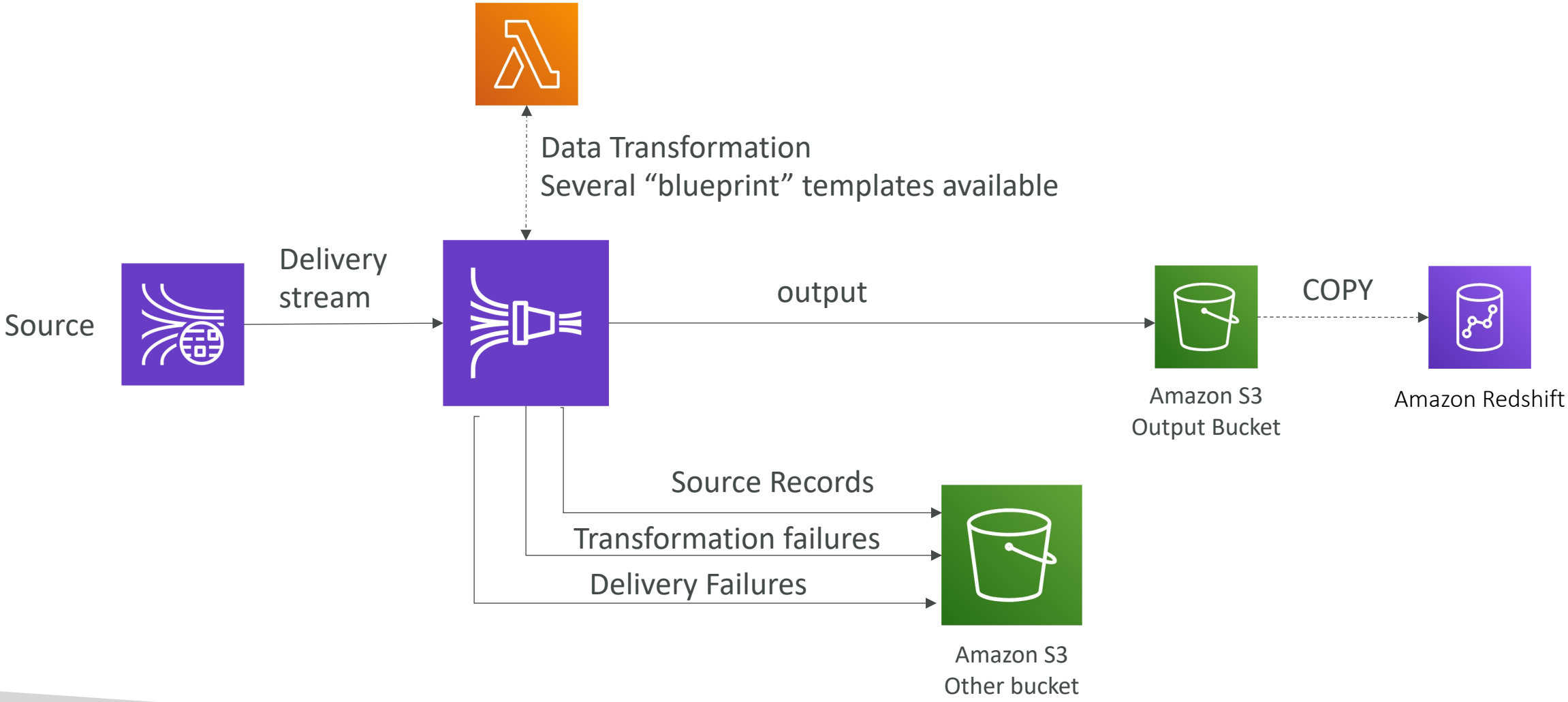


- Fully Managed Service, no administration
- **Near Real Time** (**Buffer based on time and size, optionally can be disabled**)
- Load data into Redshift / Amazon S3 / OpenSearch / Splunk
- Automatic scaling
- Supports many data formats
- Data Conversions from JSON to Parquet / ORC (only for S3)
- Data Transformation through AWS Lambda (ex: CSV => JSON)
- Supports **compression** when target is Amazon S3 (GZIP, ZIP, and SNAPPY)
- Only GZIP is the data is further loaded into Redshift
- Pay for the amount of data going through Firehose
- Spark / KCL do *not* read from KDF

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Kinesis Data Firehose Delivery Diagram



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Firehose Buffer Sizing

- Firehose accumulates records in a buffer
- The buffer is flushed based on time and size rules

- Buffer Size (ex: 32MB): if that buffer size is reached, it's flushed
- Buffer Time (ex: 2 minutes): if that time is reached, it's flushed
- Firehose can automatically increase the buffer size to increase throughput

- High throughput => Buffer Size will be hit
- Low throughput => Buffer Time will be hit

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Kinesis Data Streams vs Firehose

- Streams
  - Going to write custom code (producer / consumer)
  - Real time (~200 ms latency for classic, ~70 ms latency for enhanced fan-out)
  - On-demand or Provisioned mode
  - Data Storage for 1 to 365 days, replay capability, multi consumers
  - Use with Lambda to insert data in real-time to OpenSearch (for example)

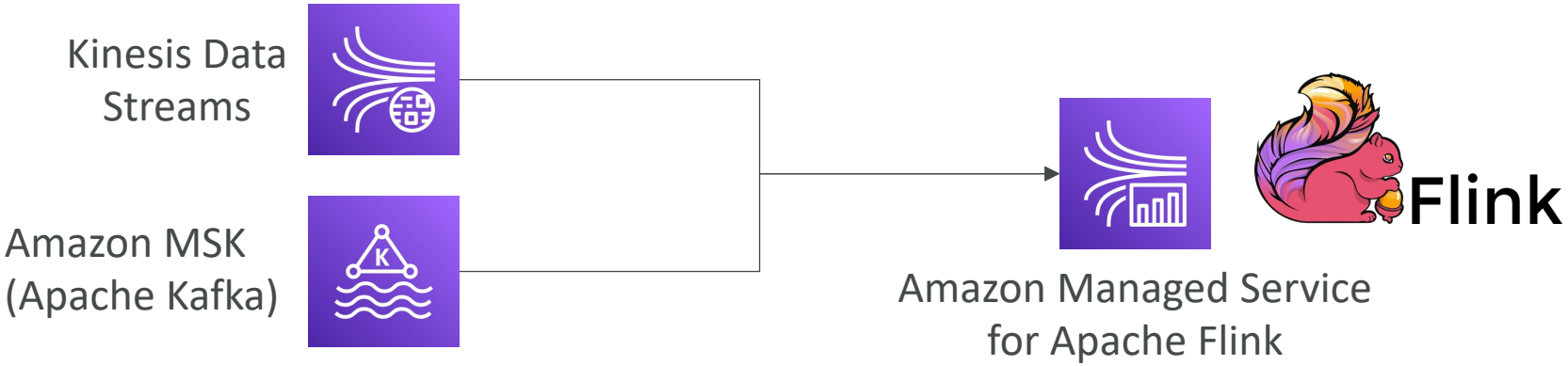
- Firehose
  - Fully managed, send to S3, Splunk, Redshift, OpenSearch
  - Serverless data transformations with Lambda
  - **Near** real time
  - Automated Scaling
  - No data storage

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Amazon Managed Service for Apache Flink

- *Previously named: Kinesis Data Analytics for Apache Flink*
- Flink (Java, Scala or SQL) is a framework for processing data streams

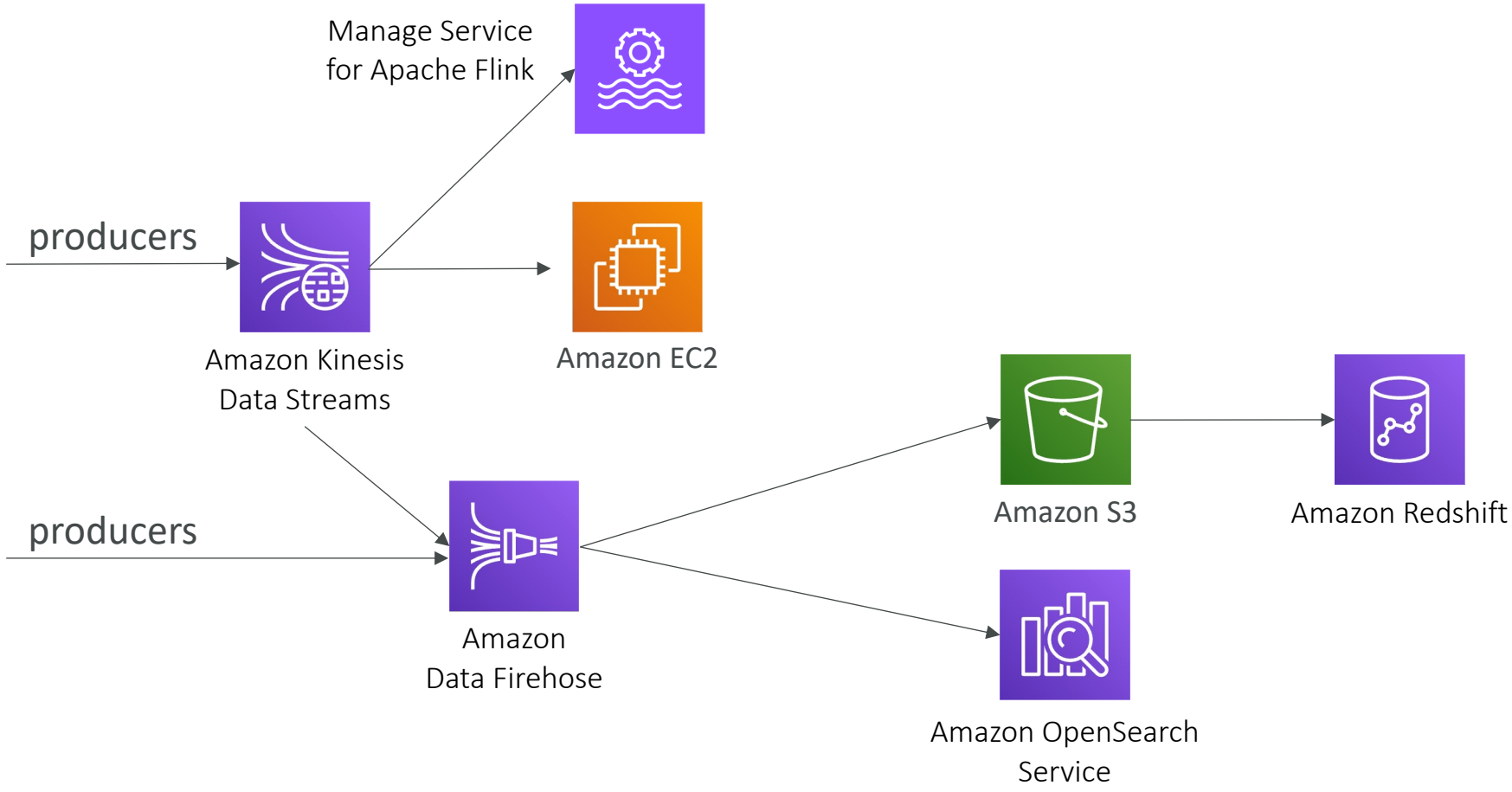


- Run any Apache Flink application on a managed cluster on AWS
  - Provisioned compute resources, parallel computation, automatic scaling
  - Application backups (implemented as checkpoints and snapshots)
  - Use any Apache Flink programming features to transform data
  - Important: Flink does not read from Amazon Data Firehose

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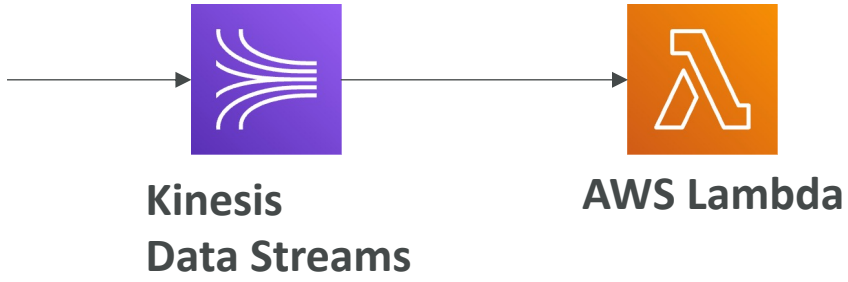
Full Data Engineering Pipeline
Real-Time Layer

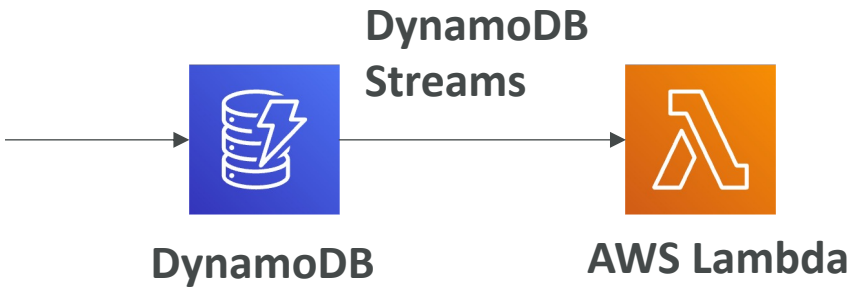


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Streaming Architectures
3000 messages of 1 KB per second





Kinesis

- 3 shards: 3MB/s in
- 3 * $0.015/hr = **$32.4/mth**
- Must use KDF for output to S3

DynamoDB + Streams

- 3000 WCU = 3 MB/s
- = **$1,450.90 / month**
- Storage in DynamoDB

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Comparison Charts

	Kinesis Data Streams	SQS	SQS FIFO	SNS	DynamoDB	S3
Data	Immutable	Immutable	Immutable	Immutable	Mutable	Mutable
Retention	1-365 days, export to S3 using KDF	1-14 days	1-14 days	No retention	Infinite or can implement TTL	Infinite, can setup lifecycle policies
Ordering	Per shard	No ordering	Per group-id	No ordering	No ordering	No ordering
Scalability	Provision shards	Soft limit	300 msg/s Or 3000 if batch	Soft limit	WCU & RCU On-demand	Infinite 3500 PUT 5500 GET / prefix
Readers	EC2, Lambda, KDF, KDA, KCL (checkpoint)	EC2, Lambda	EC2, Lambda	HTTP, Lambda, Email, SQS…	DynamoDB Streams	SDK, S3 Events
Latency	KDS (200 ms) KDF (1 min)	Low (10-100ms)	Low (10-100ms)	Low (10-100ms)	Low (10-100ms)	Low (10-100ms)

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Amazon Managed Streaming for Apache Kafka (Amazon MSK)

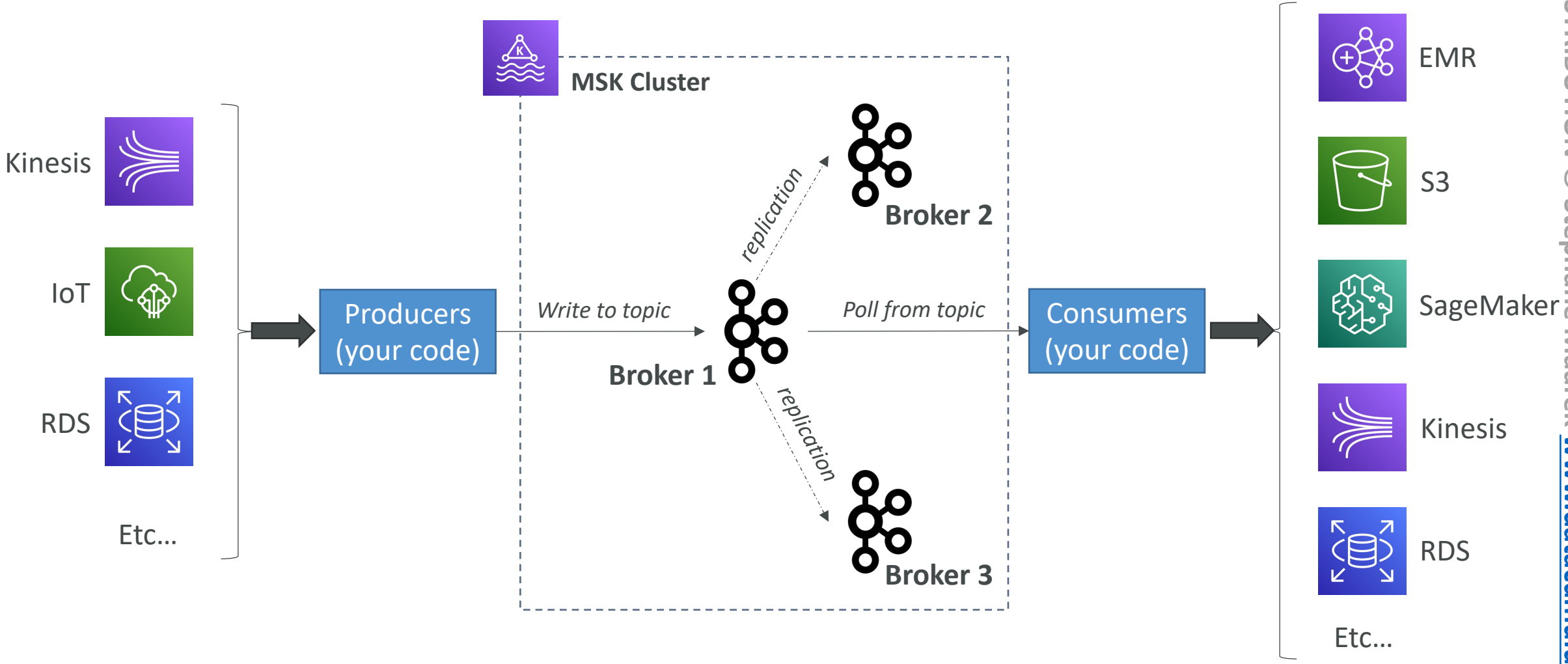


- Alternative to Amazon Kinesis
- Fully managed Apache Kafka on AWS
  - Allow you to create, update, delete clusters
  - MSK creates & manages Kafka brokers nodes & Zookeeper nodes for you
  - Deploy the MSK cluster in your VPC, multi-AZ (up to 3 for HA)
  - Automatic recovery from common Apache Kafka failures
  - Data is stored on EBS volumes **for as long as you want**
- **MSK Serverless**
  - Run Apache Kafka on MSK without managing the capacity
  - MSK automatically provisions resources and scales compute & storage

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Apache Kafka at a high level



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Kinesis Data Streams vs. Amazon MSK



Kinesis Data Streams

- 1 MB message size limit
- Data Streams with Shards
- Shard Splitting & Merging
- TLS In-flight encryption
- KMS at-rest encryption



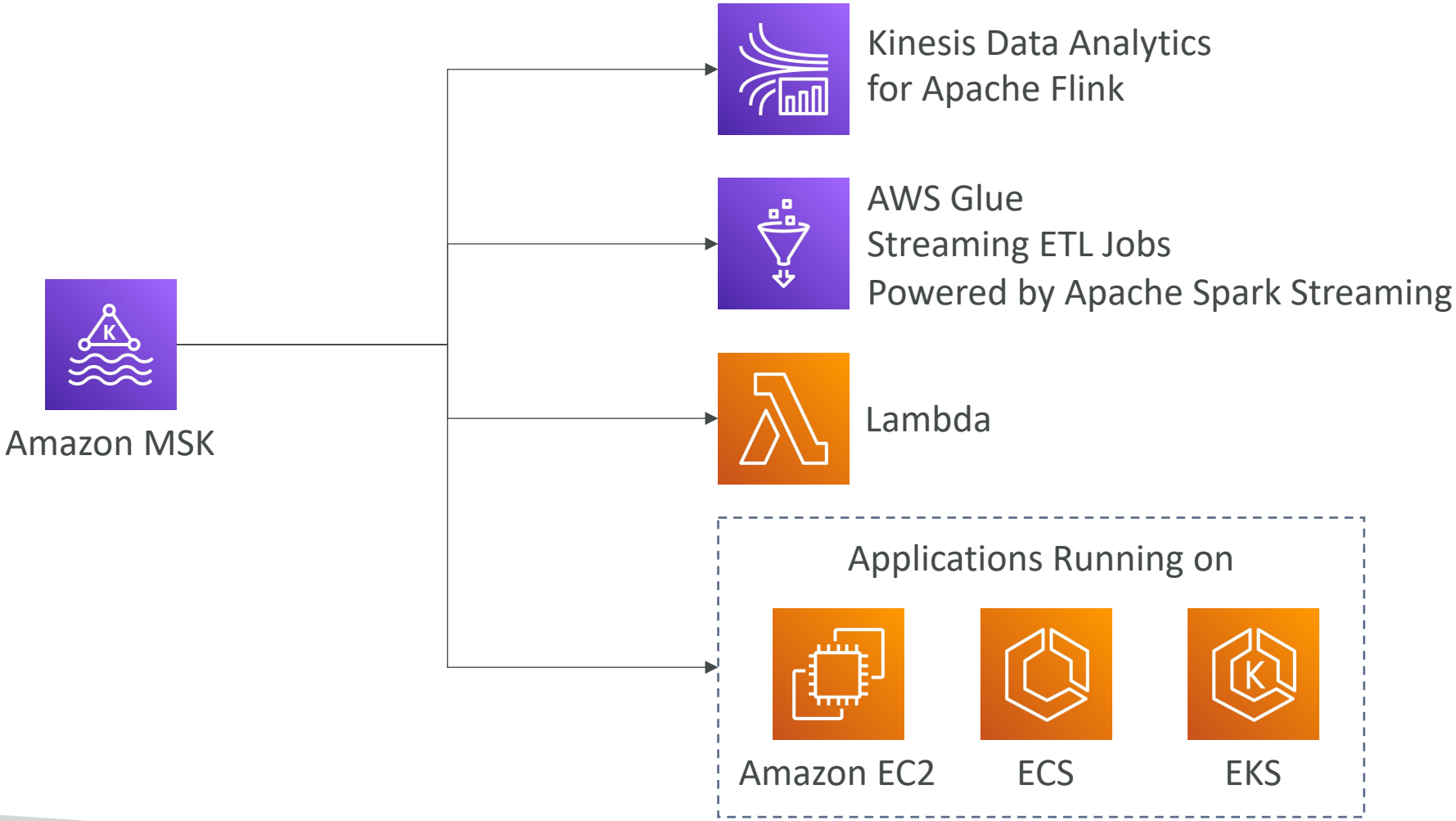
Amazon MSK

- 1MB default, configure for higher (ex: 10MB)
- Kafka Topics with Partitions
- Can only add partitions to a topic
- PLAINTEXT or TLS In-flight Encryption
- KMS at-rest encryption

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Amazon MSK Consumers



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AWS Batch

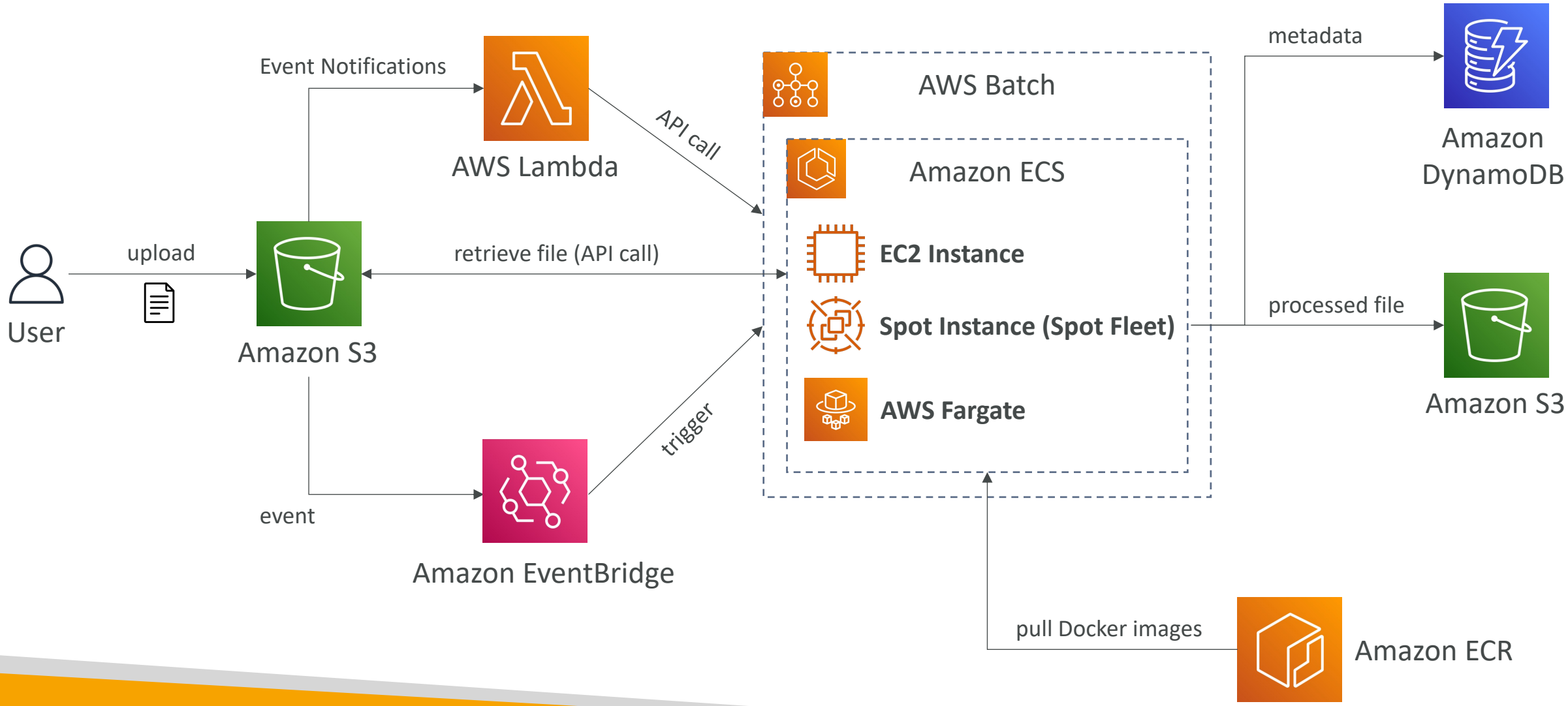


- Run batch jobs as Docker images
- Two options:
  1. **Managed Compute Environment**
     - Dynamic provisioning of the instances (EC2 & **Spot Instances**) – in VPC
     - AWS Fargate (full serverless offering)
  2. **Unmanaged Compute Environment** (ECS & EKS)
- Optimal quantity and type based on volume and requirements
- You just pay for the underlying resources used
- Example: batch process of images, running thousands of concurrent jobs
- Schedule Batch Jobs using Amazon EventBridge
- Orchestrate Batch Jobs using AWS Step Functions

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AWS Batch – Solution Architecture



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Batch vs. Lambda

- Lambda:
  - Time limit
  - Limited runtimes (built in runtimes, or Docker images built for Lambda)
  - Limited temporary disk space
  - Serverless

- Batch:
  - No time limit
  - Any runtime as long as it's packaged as a Docker image
  - Rely on EBS / instance store for disk space
  - Relies on EC2 (can be managed by AWS) or AWS Fargate

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AWS Batch – Compute Environments

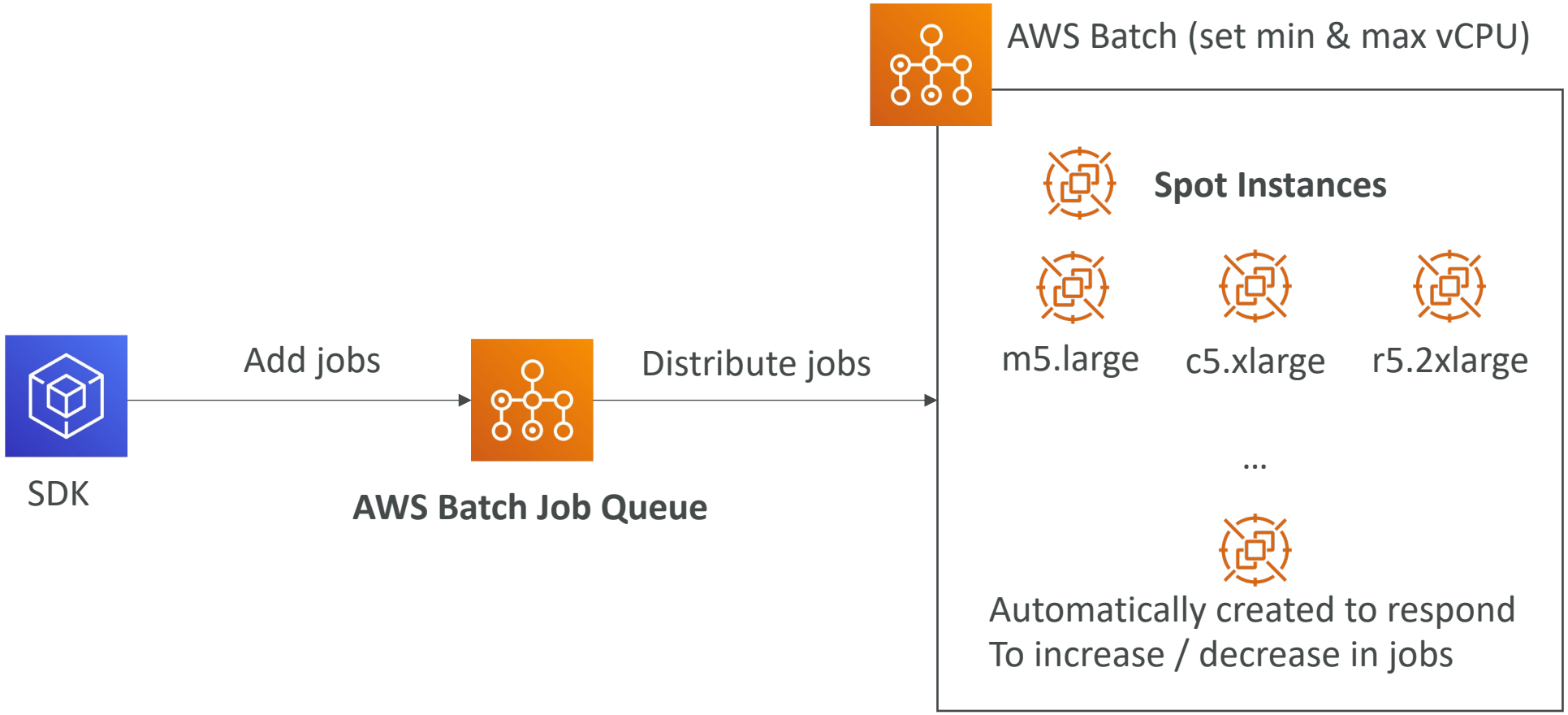
- Managed Compute Environment:
  - AWS Batch managed the capacity and instance types within the environment
  - **You can choose EC2 On-Demand or Spot Instances**
  - **You can choose Fargate On-Demand or Fargate Spot Instances**
  - You can set a maximum price for Spot Instances
  - Launched within your own VPC
    - If you launch within your own private subnet, make sure it has access to the ECS service
    - Either using a NAT gateway / instance or using VPC Endpoints for ECS

- Unmanaged Compute Environment (EC2, ECS, EKS)
  - You control and manage your compute resources, provisioning and scaling

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AWS Batch – Managed Compute Environment

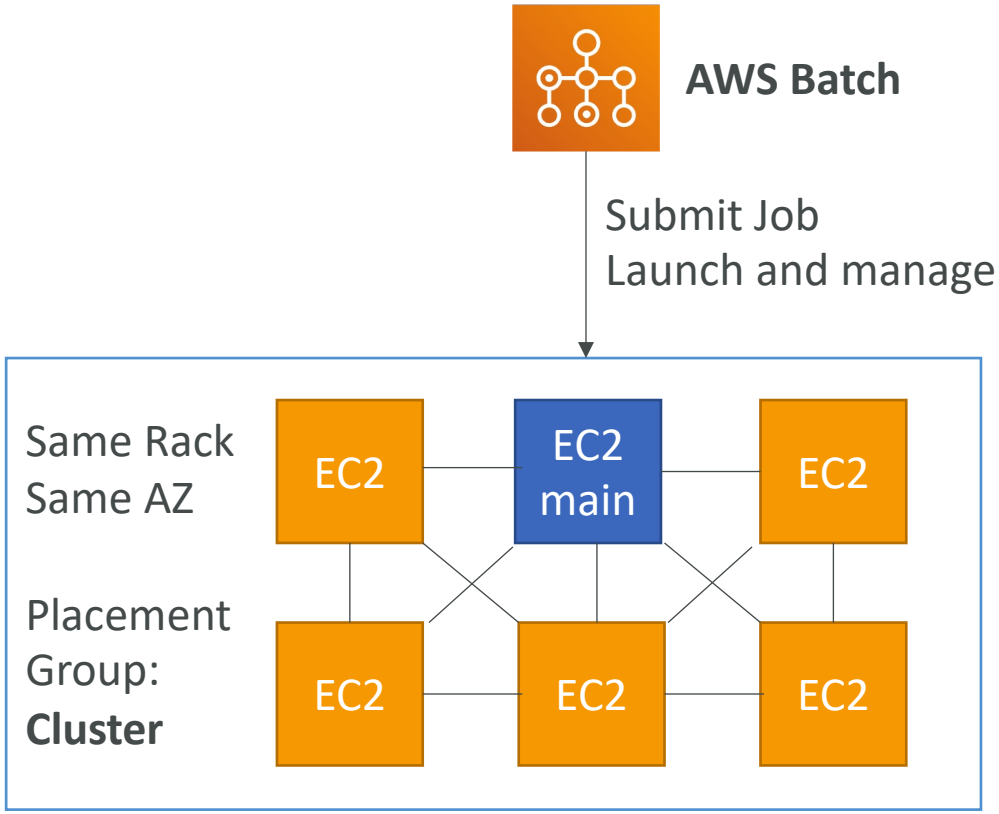


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AWS Batch – Multi Node Mode

- **Multi Node:** large scale, good for HPC (high performance computing)
  - Leverages multiple EC2 / ECS instances at the same time
  - Good for tightly coupled workloads
  - Represents a single job, and specified how many nodes to create for the job
  - 1 main node, and many child node.
  - **Does not work with Spot Instances**
  - Works better if your EC2 launch mode is a placement group "cluster"



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Amazon EMR

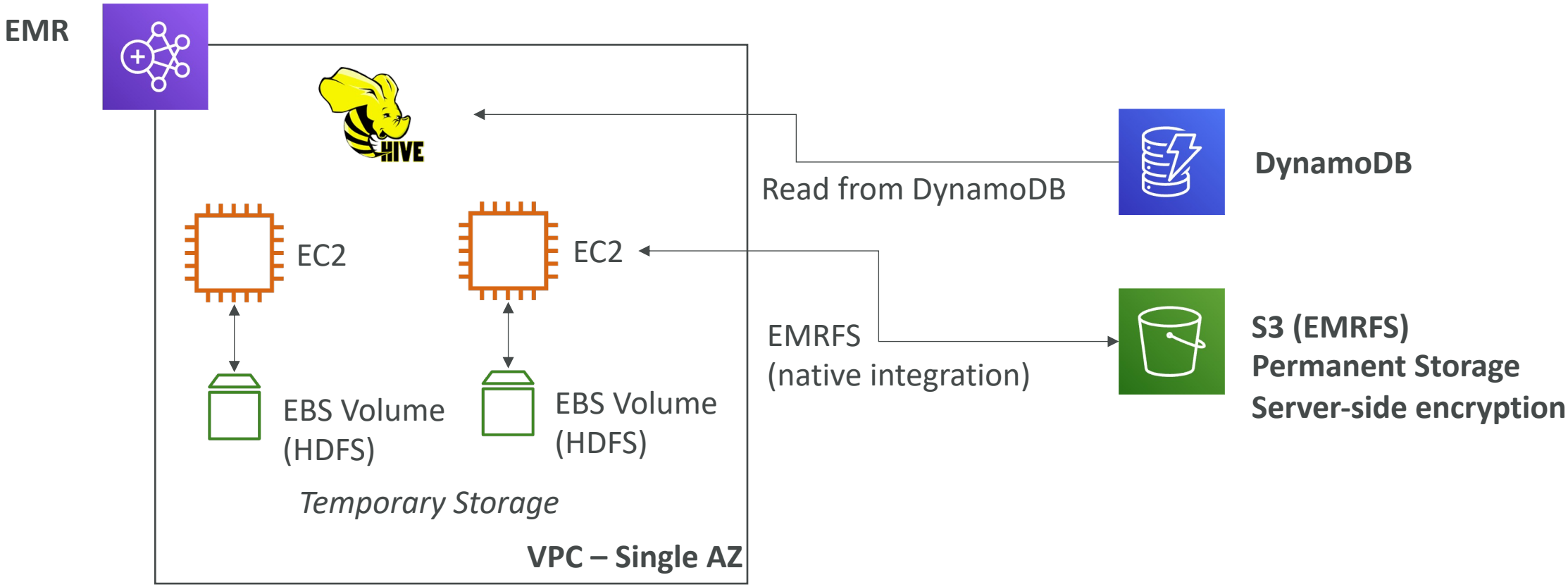


- EMR stands for "Elastic MapReduce"
- EMR helps creating **Hadoop clusters (Big Data)** to analyze and process vast amount of data
- The clusters can be made of hundreds of EC2 instances
- Also supports Apache Spark, HBase, Presto, Flink…
- EMR takes care of all the provisioning and configuration of EC2
- Auto-scaling with CloudWatch
- Use cases: data processing, machine learning, web indexing, big data…

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EMR – Integrations



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Amazon EMR – Node types & purchasing

- **Master Node:** Manage the cluster, coordinate, manage health – long running
- **Core Node:** Run tasks and store data – long running
- **Task Node (optional):** Just to run tasks – usually Spot
- **Purchasing options:**
  - On-demand: reliable, predictable, won't be terminated
  - Reserved (min 1 year): cost savings (EMR will automatically use if available)
  - Spot Instances: cheaper, can be terminated, less reliable

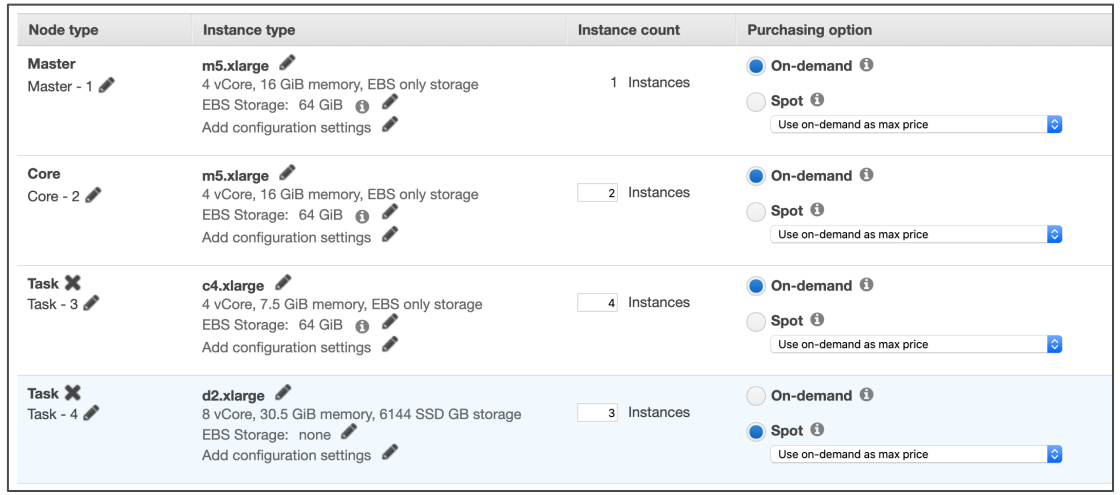
- Can have long-running cluster, or transient (temporary) cluster

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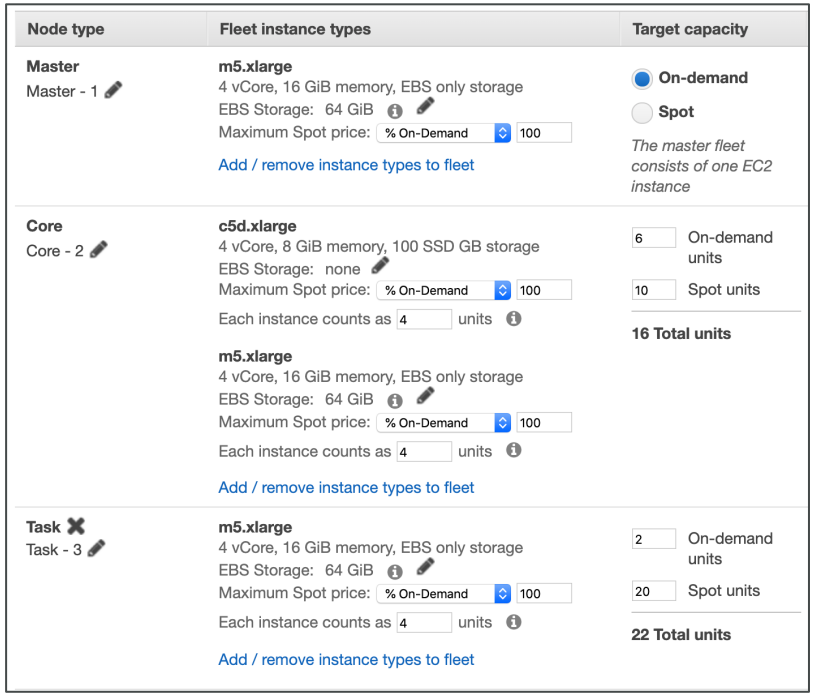
Amazon EMR – Instance Configuration

- **Uniform instance groups:** select a single instance type and purchasing option for each node



Node type	Instance type	Instance count	Purchasing option
Master Master - 1	**m5.xlarge** 4 vCore, 16 GiB memory, EBS only storage EBS Storage: 64 GiB Add configuration settings	1 Instances	On-demand Spot Use on-demand as max price
Core Core - 2	**m5.xlarge** 4 vCore, 16 GiB memory, EBS only storage EBS Storage: 64 GiB Add configuration settings	2 Instances	On-demand Spot Use on-demand as max price
Task Task - 3	**c4.xlarge** 4 vCore, 7.5 GiB memory, EBS only storage EBS Storage: 64 GiB Add configuration settings	4 Instances	On-demand Spot Use on-demand as max price
Task Task - 4	**d2.xlarge** 8 vCore, 30.5 GiB memory, 6144 SSD GB storage EBS Storage: none Add configuration settings	3 Instances	On-demand Spot Use on-demand as max price

- **Instance fleet:** select target capacity, mix instance types and purchasing options



Node type	Fleet instance types	Target capacity
Master Master - 1	**m5.xlarge** 4 vCore, 16 GiB memory, EBS only storage EBS Storage: 64 GiB Maximum Spot price: % On-Demand 100 Add / remove instance types to fleet	On-demand Spot *The master fleet consists of one EC2 instance*
Core Core - 2	**c5d.xlarge** 4 vCore, 8 GiB memory, 100 SSD GB storage EBS Storage: none Maximum Spot price: % On-Demand 100 Each instance counts as 4 units **m5.xlarge** 4 vCore, 16 GiB memory, EBS only storage EBS Storage: 64 GiB Maximum Spot price: % On-Demand 100 Each instance counts as 4 units Add / remove instance types to fleet	6 On-demand units 10 Spot units **16 Total units**
Task Task - 3	**m5.xlarge** 4 vCore, 16 GiB memory, EBS only storage EBS Storage: 64 GiB Maximum Spot price: % On-Demand 100 Each instance counts as 4 units Add / remove instance types to fleet	2 On-demand units 20 Spot units **22 Total units**

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Running Jobs on AWS

Strategy 1: Provision EC2 instance (long running - CRON jobs)



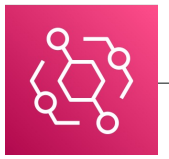
EC2

Strategy 3: Reactive Workflow

EventBridge
S3 Events
API Gateway
SQS, SNS
Etc…



Strategy 5: use Fargate



EventBridge



Fargate

Strategy 2: Amazon EventBridge + Lambda (cron)



cron schedule

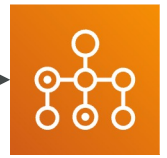


EventBridge

Strategy 4: use AWS Batch



EventBridge



Batch

Strategy 6: Use EMR (step execution or cluster)



EMR

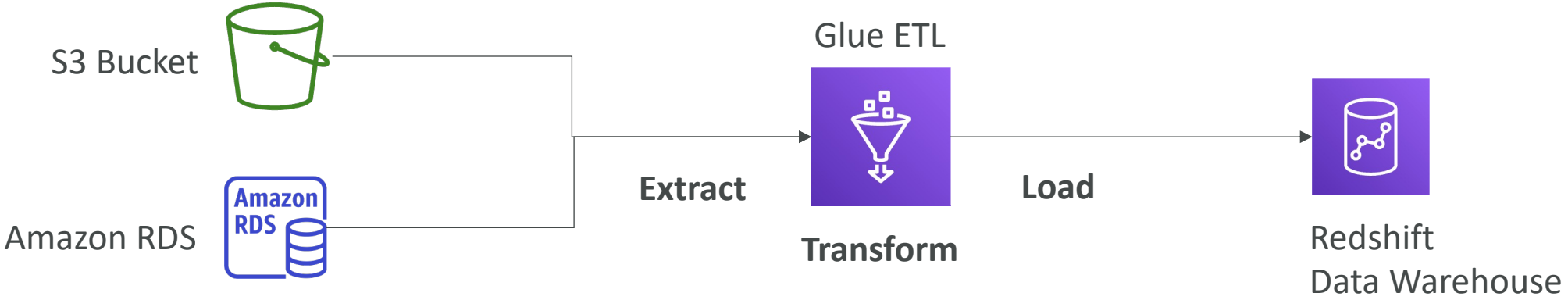
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AWS Glue



- Managed **extract, transform, and load (ETL)** service
- Useful to prepare and transform data for analytics
- Fully **serverless** service



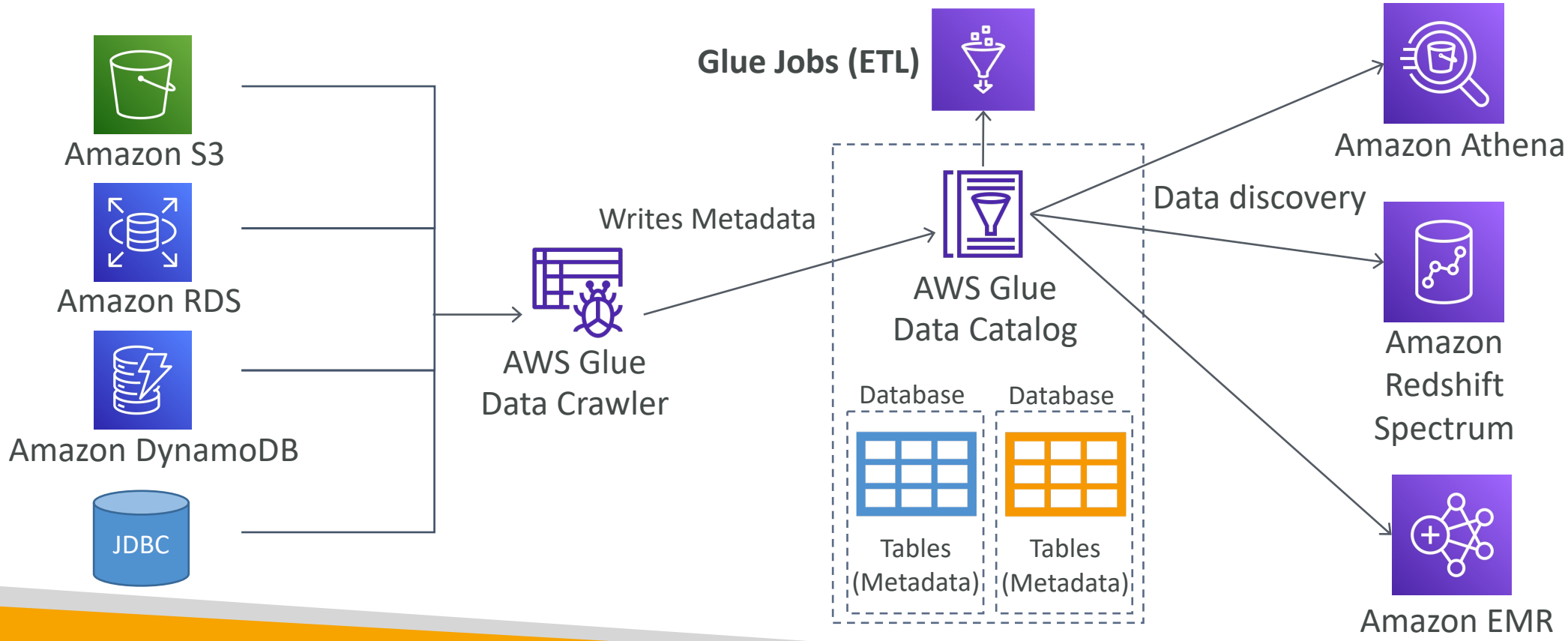
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Glue Data Catalog



- Glue Data Catalog: catalog of datasets



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Redshift Overview



- Redshift is based on PostgreSQL, but **it's not used for OLTP**
- **It's OLAP – online analytical processing (analytics and data warehousing)**
- 10x better performance than other data warehouses, scale to PBs of data
- **Columnar** storage of data (instead of row based)
- Massively Parallel Query Execution (MPP)
- Two modes: Provisioned cluster or Serverless cluster
- Has a SQL interface for performing the queries
- BI tools such as AWS Quicksight or Tableau integrate with it

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Redshift Continued…



- Data is loaded from **S3, Kinesis Firehose, DynamoDB, DMS…**
- Based on node type: up to 100+ nodes, up to 16 TB of space per node
- Can provision multiple nodes, with **Multi-AZ only for some clusters**
- Leader node: for query planning, results aggregation
- Compute node: for performing the queries, send results to leader
- Backup & Restore, Security VPC / IAM / KMS, Monitoring
- Redshift Enhanced VPC Routing: COPY / UNLOAD goes through VPC
- **Redshift is provisioned, so it's worth it when you have a sustained usage (use Athena if the queries are sporadic instead)**

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Redshift – Snapshots & DR

- Snapshots are point-in-time backups of a cluster, stored internally in S3
- Snapshots are incremental (only what has changed is saved)
- You can restore a snapshot into a **new cluster**
- **Automated: every 8 hours, every 5 GB, or on a schedule. Set retention**
- Manual: snapshot is retained until you delete it

- You can configure Amazon Redshift to automatically copy snapshots (automated or manual) of a cluster to another AWS Region

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Cross-Region Snapshot Copy for an KMS-Encrypted Redshift



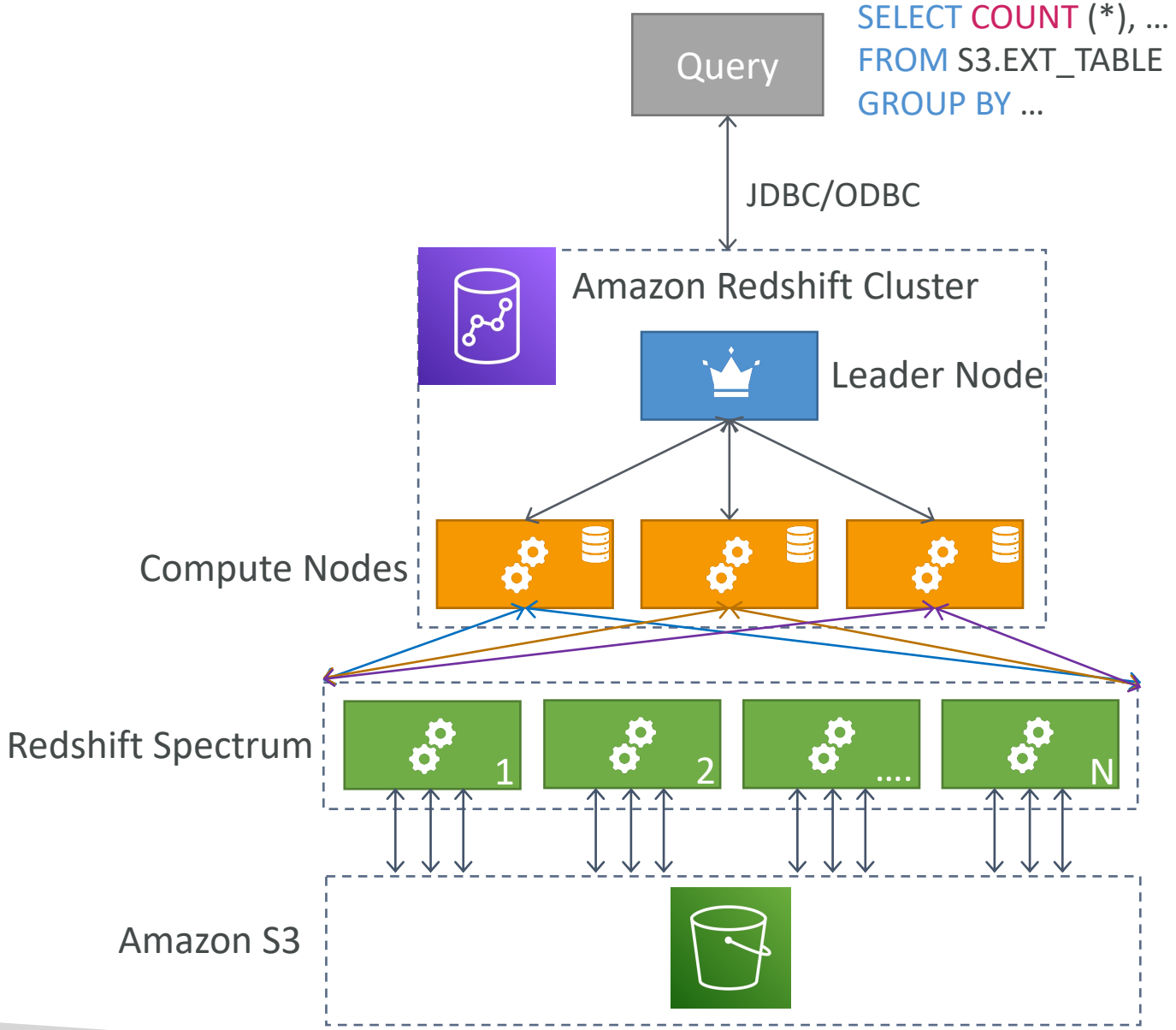
Enables Redshift to perform encryption operations in the destination Region

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Redshift Spectrum

- Query data that is already in S3 without loading it
- **Must have a Redshift cluster available to start the query**
- The query is then submitted to thousands of Redshift Spectrum nodes

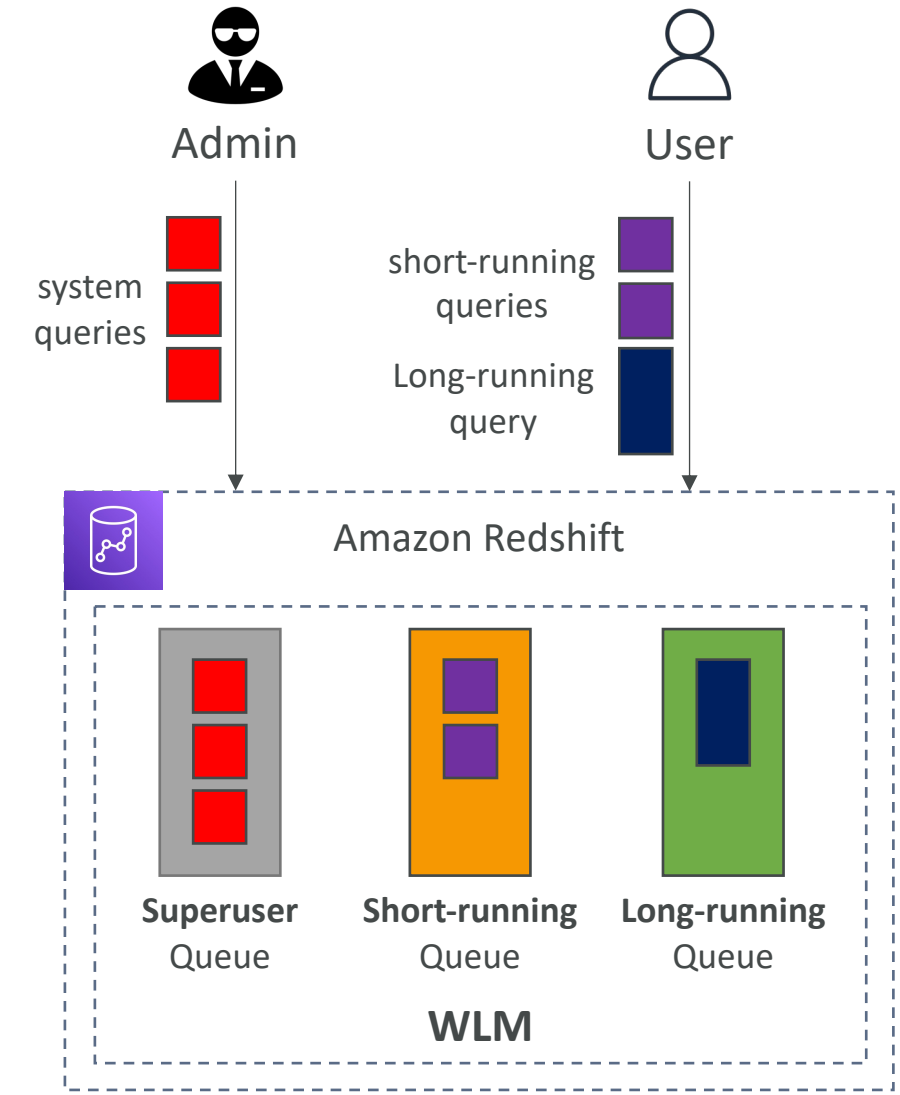


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Redshift Workload Management (WLM)

- Enables you to flexibly manage queries' priorities within workloads
- Example: prevent short, fast-running queries from getting stuck behind long-running queries
- Define multiple query queues (Superuser queue, User-defined queues)
- Route queries to the appropriate queues at runtime
- **Automatic WLM** – queues and resources managed by Redshift
- **Manual WLM** – queues and resources managed by you

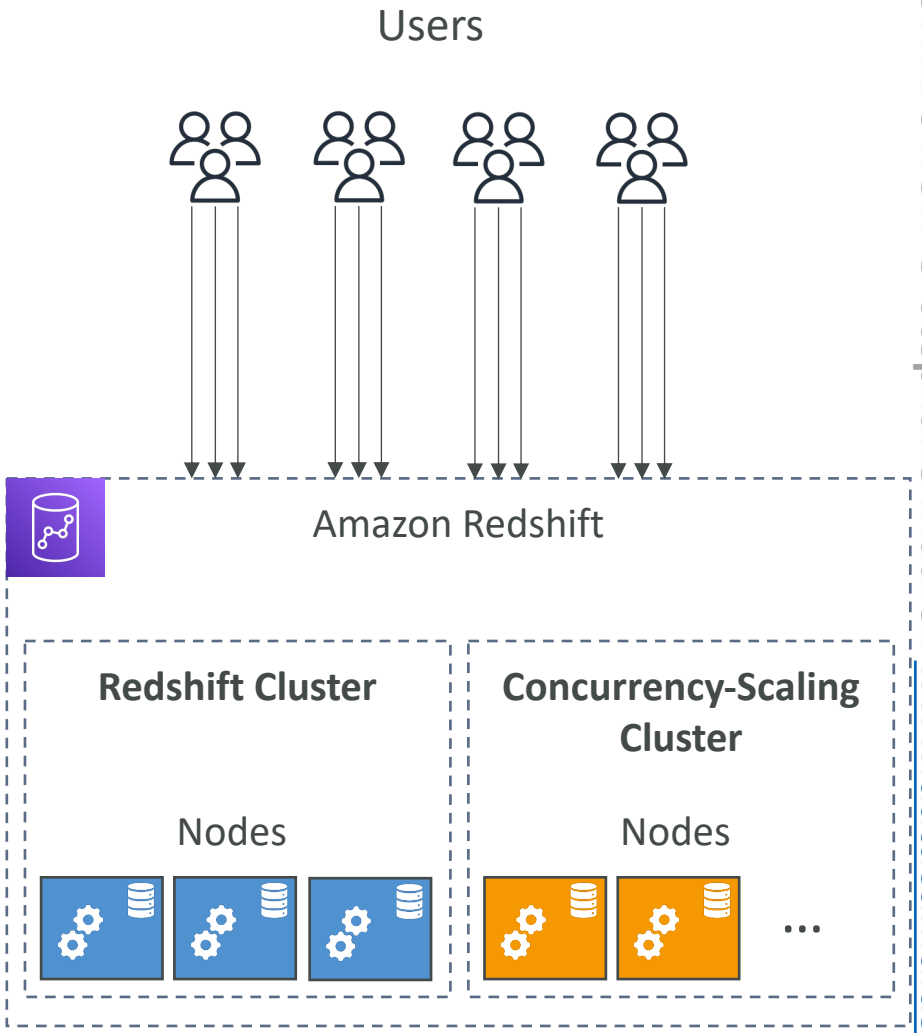


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Redshift Concurrency Scaling

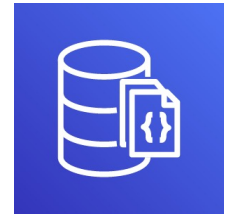
- Enables you to provide consistently fast performance with virtually unlimited concurrent users and queries
- Redshift automatically adds additional cluster capacity (**Concurrency-Scaling Cluster**) to process an increase in requests
- Ability to decide which queries sent to the concurrency-Scaling Cluster using WLM
- Charged per second



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DocumentDB



- Aurora is an "AWS-implementation" of PostgreSQL / MySQL …
- **DocumentDB is the same for MongoDB (which is a NoSQL database)**
- MongoDB is used to store, query, and index JSON data

- Similar "deployment concepts" as Aurora
- Fully Managed, highly available with replication across 3 AZ
- DocumentDB storage automatically grows in increments of 10GB

- Automatically scales to workloads with millions of requests per seconds

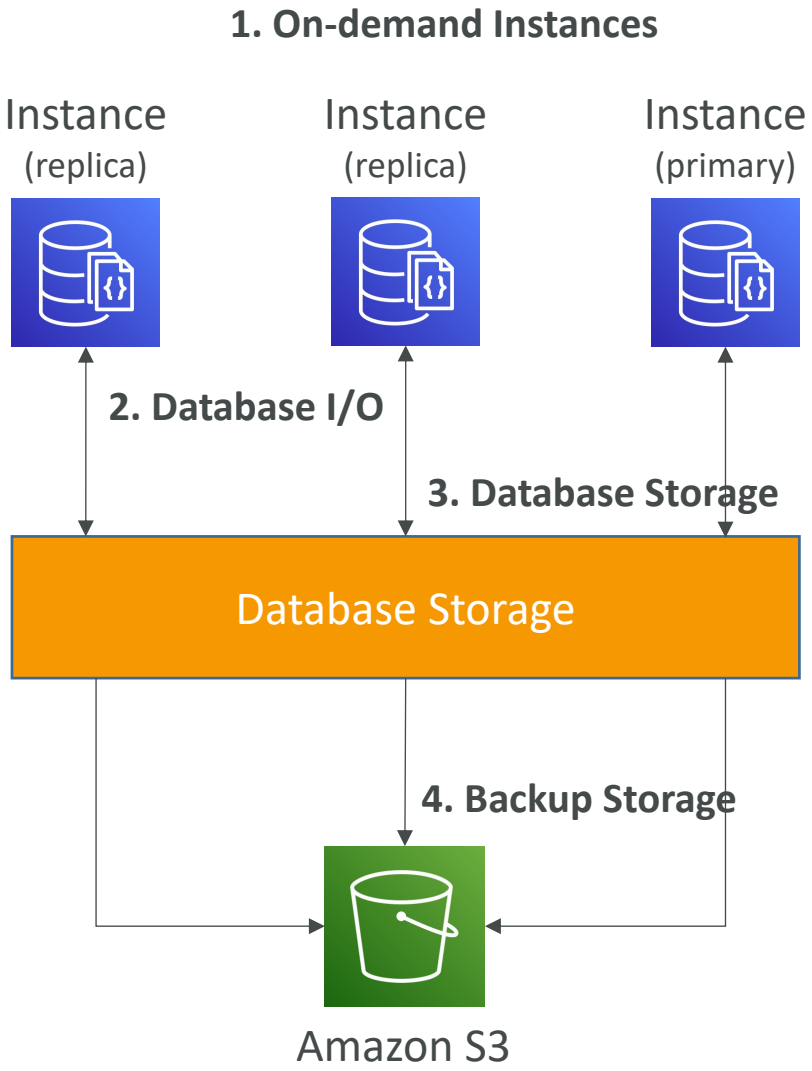
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DocumentDB – Pricing

- Pay for what you use, no upfront costs

- **On-demand Instances** (per second with minimum of 10 minutes)
- **Database I/O** – amount of I/O used when read and write (per million I/Os)
- **Database Storage** (per GB/month)
- **Backup Storage** (per GB/month)



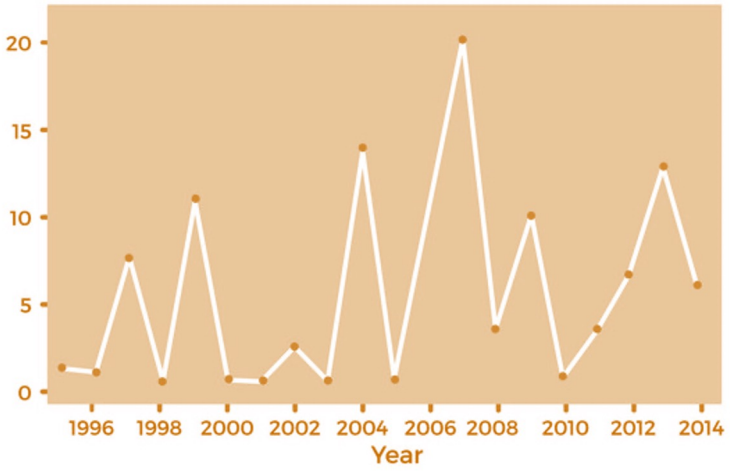
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Amazon Timestream



- Fully managed, fast, scalable, serverless **time series database**
- Automatically scales up/down to adjust capacity
- Store and analyze trillions of events per day
- 1000s times faster & 1/10th the cost of relational databases
- Scheduled queries, multi-measure records, SQL compatibility
- Data storage tiering: recent data kept in memory and historical data kept in a cost-optimized storage
- Built-in time series analytics functions (helps you identify patterns in your data in near real-time)
- Encryption in transit and at rest

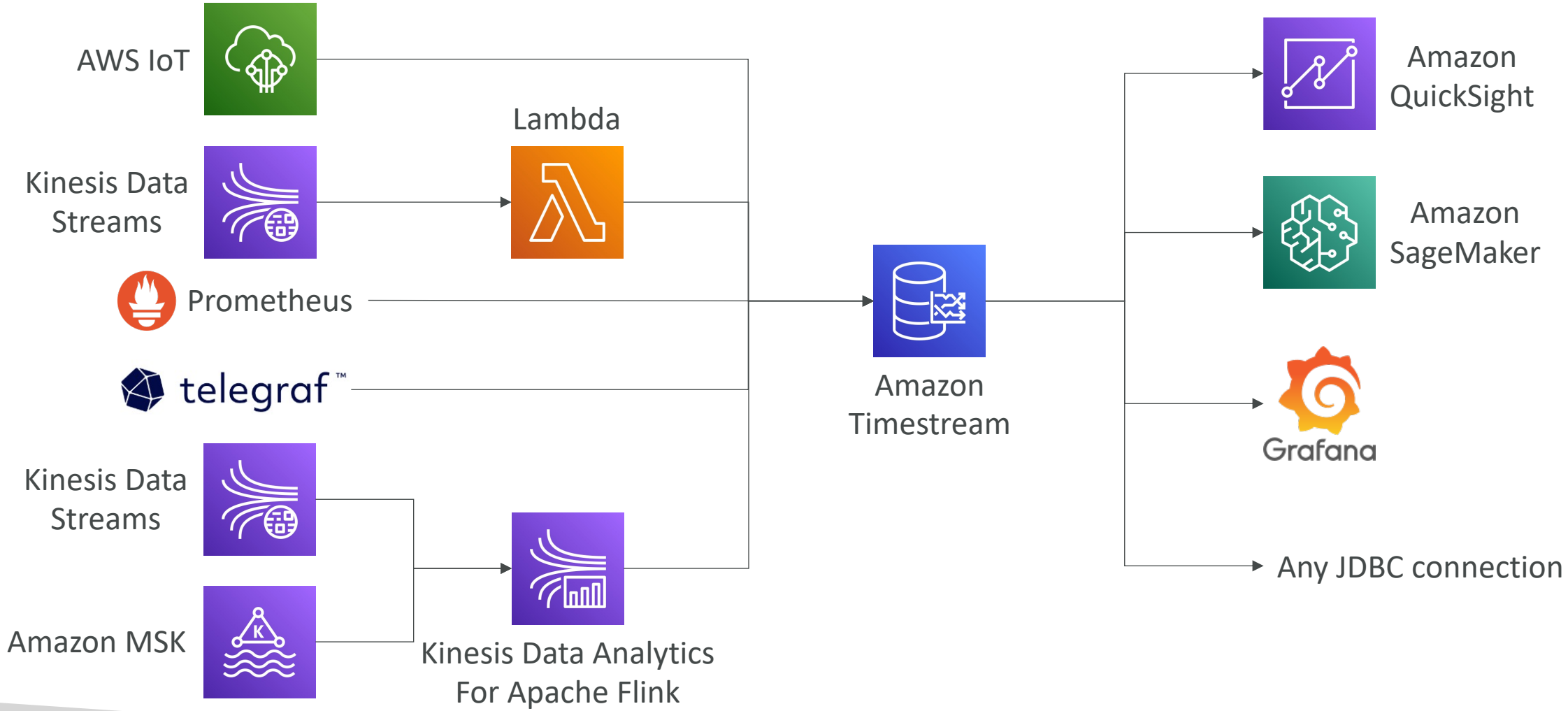


- Use cases: IoT apps, operational applications, real-time analytics, …

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Amazon Timestream – Architecture



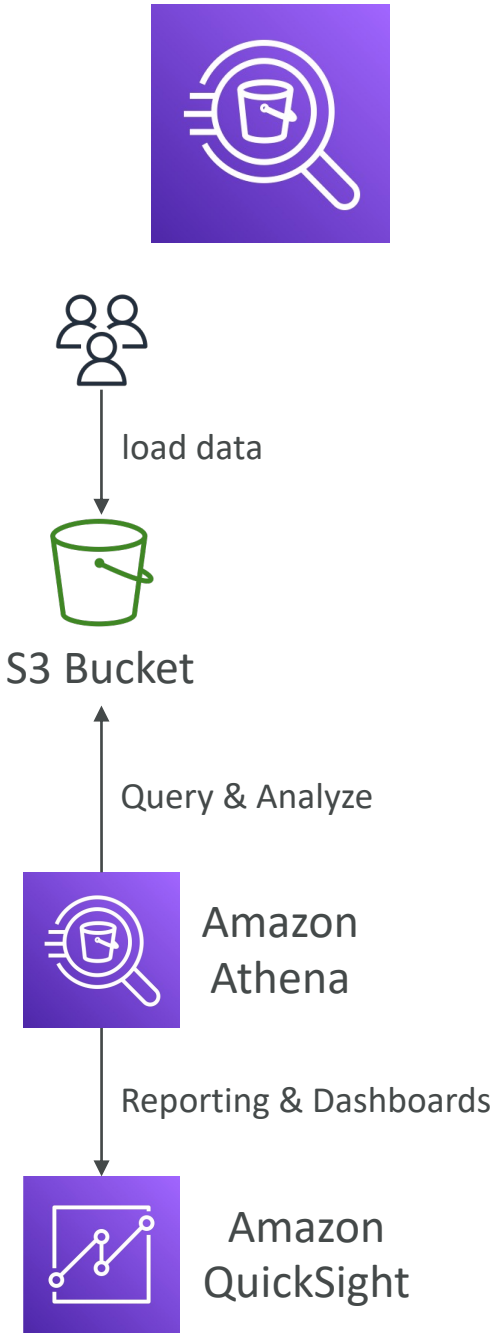
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Amazon Athena

- **Serverless** query service to analyze data stored in Amazon S3
- Uses standard SQL language to query the files (built on Presto)
- Supports CSV, JSON, ORC, Avro, and Parquet
- Pricing: $5.00 per TB of data scanned
- Commonly used with Amazon Quicksight for reporting/dashboards

- **Use cases:** Business intelligence / analytics / reporting, analyze & query VPC Flow Logs, ELB Logs, **CloudTrail trails**, etc...
- **Exam Tip:** analyze data in S3 using serverless SQL, use Athena



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Amazon Athena – Performance Improvement

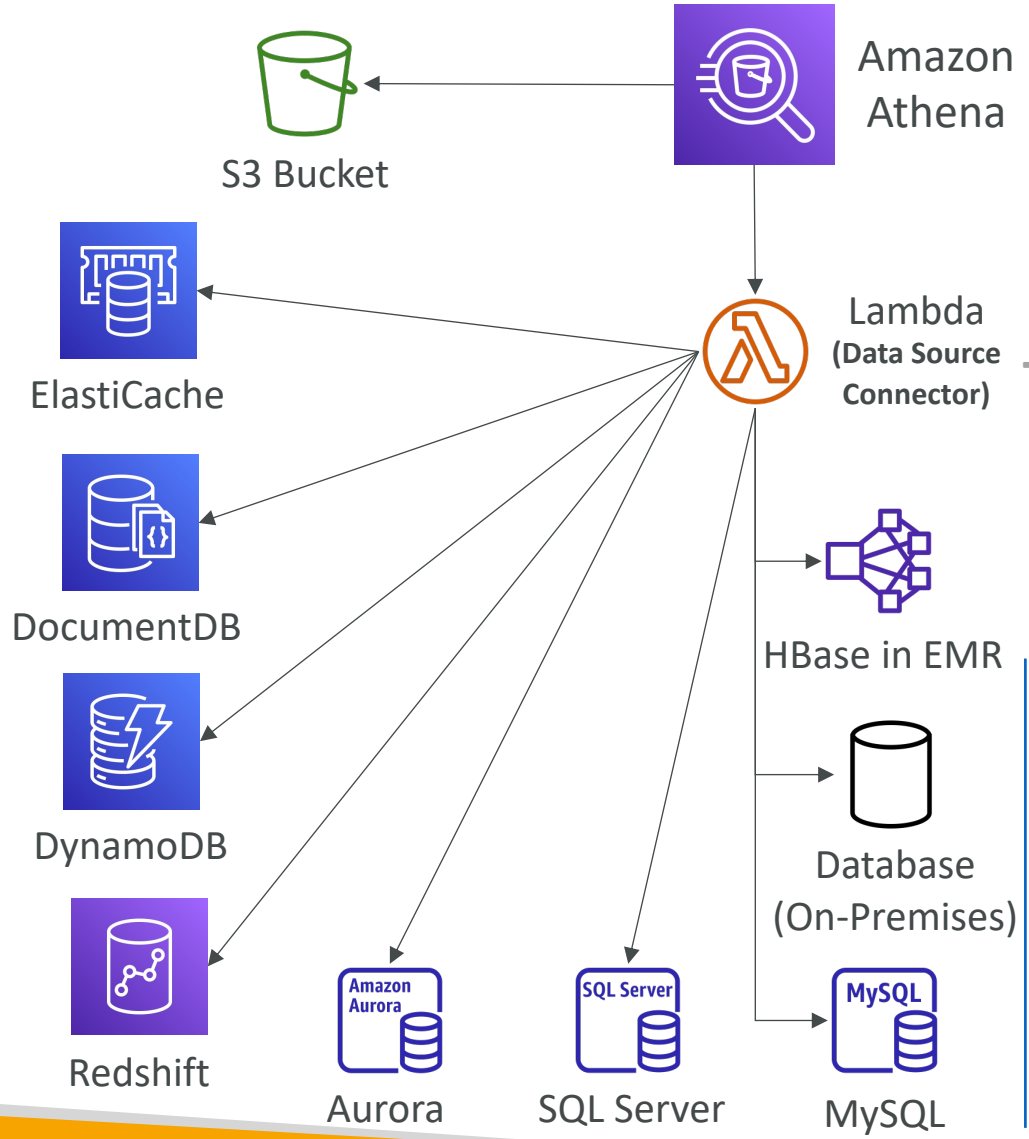
- Use **columnar data** for cost-savings (less scan)
  - Apache Parquet or ORC is recommended
  - Huge performance improvement
  - Use Glue to convert your data to Parquet or ORC
- **Compress data** for smaller retrievals (bzip2, gzip, lz4, snappy, zlip, zstd…)
- **Partition** datasets in S3 for easy querying on virtual columns
  - s3://yourBucket/pathToTable
    /<PARTITION_COLUMN_NAME>=<VALUE>
    /<PARTITION_COLUMN_NAME>=<VALUE>
    /<PARTITION_COLUMN_NAME>=<VALUE>
    /etc…
  - Example: s3://athena-examples/flight/parquet/year=1991/month=1/day=1/
- **Use larger files** (> 128 MB) to minimize overhead

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Amazon Athena – Federated Query

- Allows you to run SQL queries across data stored in relational, non-relational, object, and custom data sources (AWS or on-premises)
- Uses Data Source Connectors that run on AWS Lambda to run Federated Queries (e.g., CloudWatch Logs, DynamoDB, RDS, …)
- Store the results back in Amazon S3

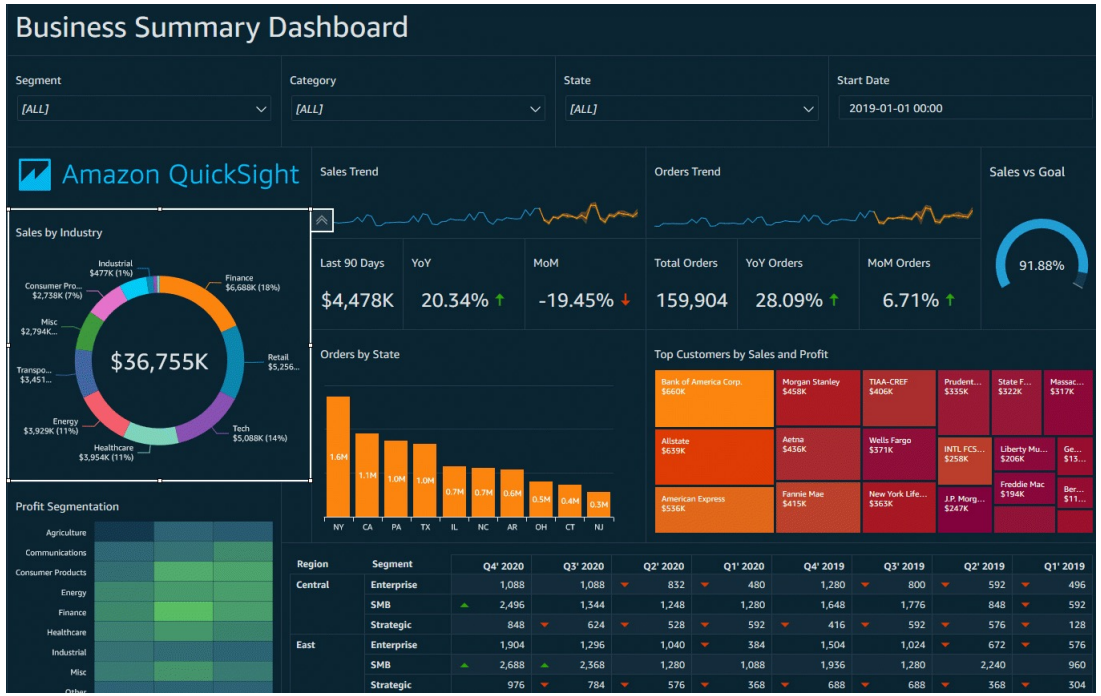


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Amazon QuickSight

- **Serverless machine learning-powered business intelligence service to create interactive dashboards**
- Fast, automatically scalable, embeddable, with per-session pricing
- Use cases:
  - Business analytics
  - Building visualizations
  - Perform ad-hoc analysis
  - Get business insights using data
- Integrated with RDS, Aurora, Athena, Redshift, S3…
- **In-memory computation using SPICE** engine if data is imported into QuickSight
- Enterprise edition: Possibility to setup **Column-Level security (CLS)**

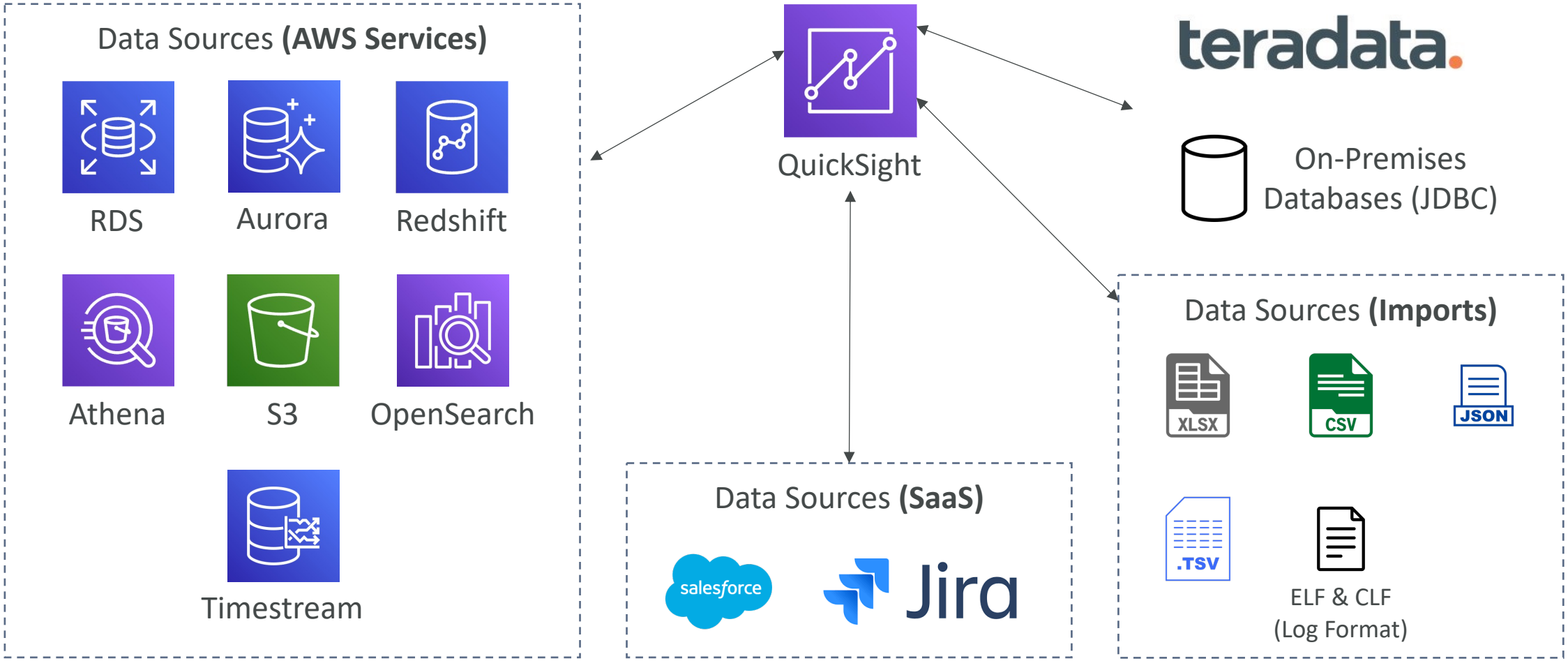


https://aws.amazon.com/quicksight/

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QuickSight Integrations



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QuickSight – Dashboard & Analysis

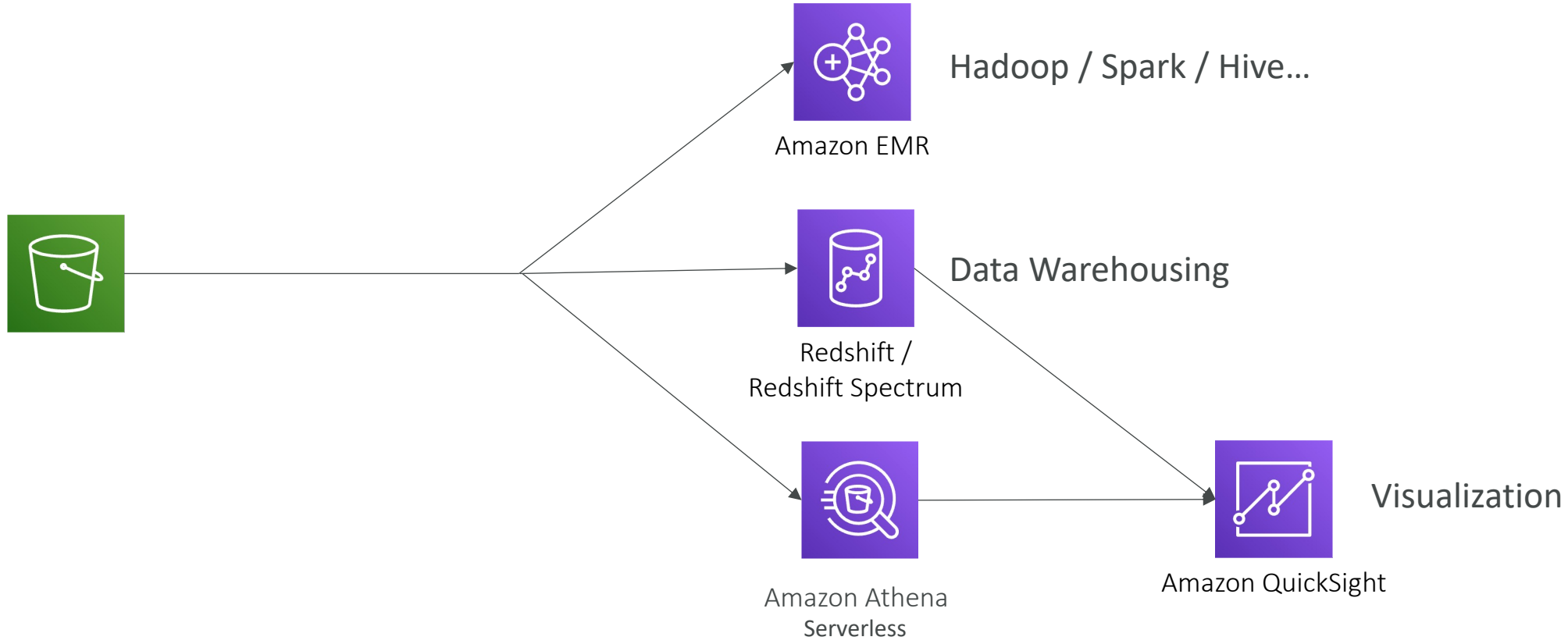
- Define Users (standard versions) and Groups (enterprise version)
  - These users & groups only exist within QuickSight, not IAM !!
- A *dashboard*…
  - is a read-only snapshot of an analysis that you can share
  - preserves the configuration of the analysis (filtering, parameters, controls, sort)

- You can share the analysis or the dashboard with Users or Groups
- To share a dashboard, you must first publish it
- Users who see the dashboard can also see the underlying data

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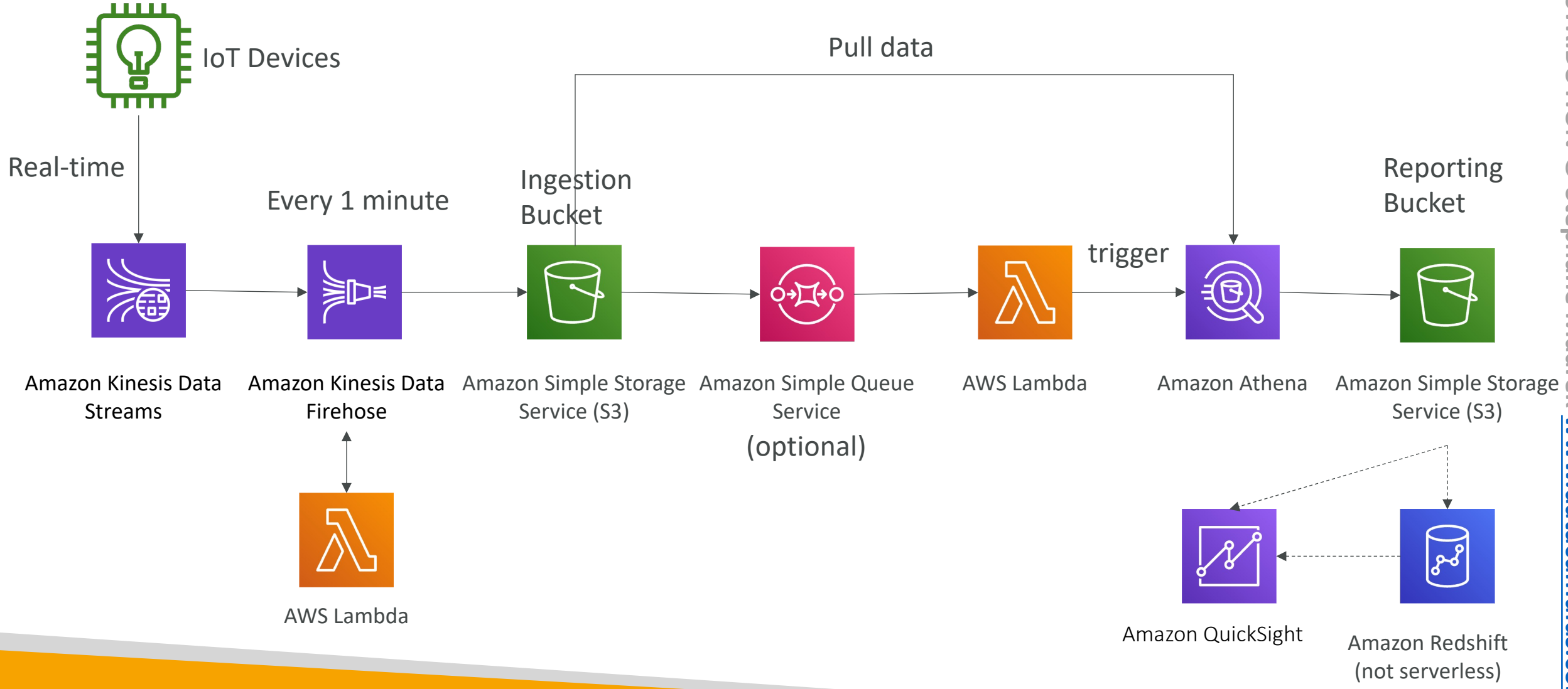
Full Data Engineering Pipeline
Analytics layer



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Big Data Ingestion Pipeline



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Comparison of warehousing technologies

- EMR
  - Need to use Big Data tools such as Apache Hive, Spark
  - One long-running cluster, many jobs, with auto-scaling, or one cluster per job?
  - Purchasing options – Spot, On Demand, Reserved Instances
  - Can access data in DynamoDB and / or S3
  - Scratch data on EBS disks (HDFS) and long term storage in S3 (EMRFS)
- Athena
  - Simple queries and aggregations, data must live in S3
  - Serverless, simple SQL queries, out-of-the-box queries for many services (cost & billing..)
  - Audit queries through CloudTrail
- Redshift
  - Advanced SQL queries, must provision servers
  - Can leverage Redshift Spectrum for serverless queries on S3

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